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Shaurya Gomber

Education

2024–2029 (Expected) **Ph.D. in Computer Science**, *University of Illinois Urbana-Champaign*, USA
GPA: 4.0/4.0; *Advisor*: Prof. Gagandeep Singh

2022–2024 **M.S. in Computer Science**, *University of Illinois Urbana-Champaign*, USA
GPA: 4.0/4.0; *Advisor*: Prof. Gagandeep Singh

2016–2020 **B.Tech. in Computer Science**, *Indian Institute of Technology Guwahati*, India
GPA: 9.66/10.0 (Institute Rank 3)

Research Interests

I am broadly interested in formal methods and automated reasoning. My current research focuses on developing precise and efficient program analyses using adaptable abstract interpretation, gradient-guided optimization, and GPU-accelerated nonlinear reasoning. I am also interested in SAT/SMT solving, mathematical optimization, and neuro-symbolic systems that combine learning with formal reasoning.

Publications

Conferences

PLDI 2026 **Evolving Abstract Transformers for Gradient-Guided, Adaptable Abstract Interpretation**
Shaurya Gomber, Debangshu Banerjee, Gagandeep Singh

ESOP 2026 **Efficient Ranking Function-Based Termination Analysis via Bidirectional Decompositional Search**
Yasmin Sarita, Avaljot Singh, Shaurya Gomber, Mahesh Viswanathan, Gagandeep Singh

Workshops & Posters

VerifAI ICLR 2026 **Unified Operational Formalism for LLM-based Theorem-proving Systems**
Avaljot Singh*, Shaurya Gomber*, Yasmin Sarita, José Meseguer, Gagandeep Singh

VerifAI ICLR 2025 **Neural Abstract Interpretation**
Shaurya Gomber, Gagandeep Singh

Thesis

MS CS 2024, UIUC **Neural Abstract Interpretation: Leveraging neural networks for automated, efficient and differentiable abstract interpretation**
Shaurya Gomber
🏆 David J. Kuck Outstanding MS Thesis Award

Awards & Fellowships

2024 **David J. Kuck Outstanding Master's Thesis Award, UIUC** Details
2024 **Richard T. Cheng Endowed Fellowship, UIUC** Details
2019 **Institute Merit Scholarship, IIT Guwahati** Details
2016 **KVPY Government of India Scholarship** Details

Work Experience

Summer 2023 **Applied Scientist Intern**, *Automated Reasoning Group @ Amazon AWS*, Santa Clara, USA

- Worked on Zelkova, an AWS tool for SMT-based reasoning over access policies.
- Applied SMTO to design efficient encodings for type-casting semantics (e.g., numeric comparisons over strings), solving ~30k previously-unsolved production queries at ~1 min/query.
- Fixed bugs and improved the oracle I/O interface in CVC5's SMTO solver.
- Tech Stack: Java, Scala, Python, SMT Solvers (Z3, CVC5)

2020-2022 **Senior Member of Technical Staff**, *D.E. Shaw & Co.*, Hyderabad, India

- Extended the firm's low-latency trading system (processing terabytes of data daily) with on-demand computation features to optimize trader workflows.
- Contributed to design discussions for core trading system components and mentored two SDE-1s.
- Tech Stack: Java, C++, React, Git, Bash, Grafana, NumPy, Matplotlib

Summer 2019 **Software Engineering Intern**, *D.E. Shaw & Co.*, Hyderabad, India

Implemented a *type-safe low-latency API* in Java to read and write on the firm's database, achieving a 60x run-time improvement in production-critical scripts, leading to a Pre-Placement Offer.

Teaching Experience

Spring '26 Teaching Assistant, CS477 Formal Software Development Methods, UIUC

Fall '23 Teaching Assistant, CS421 Programming Languages & Compilers, UIUC

Spring '23 Teaching Assistant, CS421 Programming Languages & Compilers, UIUC

Fall '22 Teaching Assistant, CS225 Data Structures & Algorithms, UIUC

Academic Service

Artifact Evaluation Committee PLDI '25, PLDI '24

Talks

Jun 2026 **Evolving Abstract Transformers for Gradient-Guided Abstract Interpretation** Slides
Programming Languages Design and Implementation (PLDI) 2026, Boulder, CO

Oct 2025 **LINC: A Neuro-symbolic Approach for Logical Reasoning** Slides
PL/FM/SE PhD Qualification Exam, UIUC

May 2025 **Multi-Network Relational Verification and Certifiable Training** Slides
CS584 Embedded System Verification, Spring 2025, UIUC

Apr 2024 **Neural Abstract Interpretation** Slides
Formal Methods Seminar, Spring 2024, UIUC

Nov 2023 **Verification and Certified Training of PINNs** Slides
CS598 Scientific Machine Learning, Fall 2023, UIUC

Nov 2023 **Satisfiability and Synthesis Modulo Oracles** Slides
Formal Methods Seminar, Fall 2023, UIUC

May 2023 **Neural Approximations of Abstract Transformers** Slides
CS477 Formal Software Development Methods, Spring 2023, UIUC

Mar 2023 **Synthesizing Abstract Transformers** Slides
Formal Methods Seminar, Spring 2023, UIUC

Nov 2022 **Monotonic Neural Networks** Slides
CS521 Trustworthy AI Systems, Fall 2022, UIUC

Mentorship

- **Peer Mentor, SAATHI Counselling Club:** Mentored CSE freshers under the Mentor-Mentee program at IIT Guwahati.
- **Placement Lectures Coordinator, IITG:** Taught Data Structures & Algorithms and managed content and scheduling for placement preparation lectures.
- **Treasurer, CSEA (2019–20):** Managed fund allocation for events of the Computer Science and Engineering Association, IIT Guwahati.

Selected Distinctions

- **Microsoft Code.Fun.Do 2019:** National finalist, top 10 of 300+ teams; built a Blockchain-based Voting System.
- **Inter IIT Tech Meet 2018:** Represented IITG in the coding hackathon at IIT Bombay.
- **ACM ICPC 2018:** Qualified for India regionals; represented IITG at Amritapuri, Kerala.
- **KVPY 2015:** AIR 178 among 1.5M candidates (top 0.01%) in IISc Bangalore's national science aptitude exam.
- **IIT JEE Advanced 2016:** AIR 902 among 1.5M candidates (top 0.06%) in India's engineering entrance exam.
- **IIT JEE Mains 2016:** AIR 2323 among 1.5M candidates (top 0.15%) in India's engineering entrance prelims.